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H_∞ Robust Control Design for the NASA AirSTAR Flight Test Vehicle

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The Airborne Subscale Transport Aircraft Research (AirSTAR) testbed at NASA Langley Research Center is a state-of-the-art system for verification and validation of experimental flight control laws. This testbed is an effective flight control resource through its flexible architecture and rapid implementation and testing cycle. Recent AirSTAR flight testing has focused on evaluating the performance and robustness of experimental control strategies, such as adaptive control. The performance and robustness properties of any experimental controller would greatly benefit from comparison to equivalent properties of a baseline controller. This paper summarizes the design, verification, flight testing, and validation of a baseline control law. The H_∞ robust control framework is used to synthesize a multivariable control architecture. Design objectives include robustness with respect to time delay and modeling uncertainty, and satisfactory pilot handling performance. Linear analysis tools are used to verify robustness, and piloted simulations are used to verify handling performance. The performance and robustness properties of the closed-loop system are then validated through flight testing.

Nomenclature

V	Airspeed, knots
α	Angle-of-attack, rad
q	Pitch rate, rad/s
θ	Pitch angle, rad
β	Angle of sideslip, rad
p	Roll rate, rad/s
r	Yaw rate, rad/s
ϕ	Bank angle, rad
δ_{elev}	Elevator deflection, deg
δ_{ail}	Aileron deflection, deg
δ_{rud}	Rudder deflection, deg
C_{m_α}	Pitch moment stiffness
C_{l_p}	Roll moment damping

I. Introduction

A major driver for innovation and research in flight control is the need to pursue improved reliability in airborne safety critical systems. Failures in safety critical systems result in catastrophic harm or loss of life. Hence, improving and certifying the safety and reliability of such aircraft systems is a priority. Adaptive control has been proposed to improve the reliability of flight control systems subject to adverse conditions,

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such as faults or subsystem failures.[1,2] Adaptive control has the potential to improve a pilot's ability to maneuver and safely land an aircraft in the presence of a major subsystem failure. However, there are significant challenges associated with adaptive control, including certifying robustness and performance, and demonstrating its benefit vis-a-vis advanced robust control. The adaptive control research area remains highly active to address certification challenges and to validate the benefits of the architecture.

An effective resource for experimentally testing flight control algorithms, including adaptive control algorithms, is the Airborne Subscale Transport Aircraft Research (AirSTAR) testbed at NASA Langley Research Center.[3–5] The primary AirSTAR flight test vehicle is a turbine powered 5.5% dynamically scaled model of a civilian transport aircraft, often referred to as the Generic Transport Model (GTM).[6] The GTM has a wing span of 7 ft, and weighs around 55 lbs. Under normal operations, it flies at an altitude of 700 to 1100 ft, with an airspeed between 70 and 85 knots. The currently used T-2 test aircraft is shown in Figure 1. Significant wind tunnel and flight testing has been performed to identify the flight dynamics of the GTM.[7] A nonlinear simulation model of the aircraft dynamics has been developed and is readily available to the research community.[8] Experimental control algorithms are easily embedded in this simulation model for verification prior to flight testing. Hence, the AirSTAR testbed is a highly effective for experimental flight control research through its flexible architecture and rapid implementation and testing cycle.



Figure 1. The GTM T-2 test vehicle in flight.

Recent AirSTAR flight testing has focused on evaluating the performance and robustness of various adaptive control architectures.[9,10] Handling qualities and stability properties were determined for the nominal case and during adverse or failure scenarios. Commonly tested scenarios include additional time delay in the control loop and stability degradation in the longitudinal and lateral axis dynamics. To properly quantify the benefit of experimental control architectures, performance and robustness properties must be compared to those of a baseline controller. Further, the baseline controller must represent the current state-of-the art in terms of performance and robustness and be regarded as a benchmark. The primary goal of the research summarized in this paper is the design, verification, flight testing, and validation of such a baseline controller.

The control design presented is based on the H_∞ robust control framework, which results in a multivariable control architecture. Design objectives include robustness with respect to time delay and modeling uncertainty, and satisfactory pilot handling performance. The H_∞ framework balances performance and robustness properties to synthesize an optimal controller. The controller is linear, hence its robustness can be verified using standard linear analysis. Piloted simulations are used to verify handling qualities performance. The performance and robustness properties of the closed-loop system are validated through flight testing.

The remainder of the paper is structured through the following sections: Section II analyzes the aircraft flight dynamics and details the control design. Section III presents the flight test results and validation analysis. Finally, Section IV offers concluding remarks and future directions in research.

II. Aircraft Dynamics and Control Design

The GTM nonlinear simulation model is trimmed to steady level flight at 80 knots and linearized to obtain a linear model of the aircraft dynamics. A twelve state, full-order linear model is generated for the given flight condition. The four states governing longitude, latitude, altitude, and heading are eliminated as they have no impact on the flight dynamics. The remaining eight state, reduced-order model fully describes the longitudinal, lateral, and directional dynamics of the aircraft. Coupling between the longitudinal and lateral/directional axes is limited, and can be neglected for the purpose of control design. Hence, the longitudinal controller is designed independently of the lateral/directional controller. The longitudinal dynamics and control design are described in Subsection A, and the lateral/directional dynamics and control design in Subsection B. The full control design is verified with time domain simulations in Subsection C.

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A. Longitudinal Aircraft Dynamics and Control Design

A four state model, decoupled from the eight state reduced-order model, captures the primary longitudinal flight dynamics of the GTM. The states of the model are airspeed V [knots], angle-of-attack α [rad], pitch rate q [rad/s], and pitch angle θ [rad]. The control input to the model is elevator deflection δ_{elev} [deg]. The longitudinal state-space model of the GTM at 80 knots is given by the system in Equation 1.

$$\frac{d}{dt}\begin{bmatrix} V \\ \alpha \\ q \\ \theta \end{bmatrix} = \begin{bmatrix} -0.0665 & -11.4608 & 0.1439 & -32.1740 \\ -0.0035 & -2.4714 & 0.9514 & 0 \\ -0.0090 & -43.9070 & -3.4738 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} V \\ \alpha \\ q \\ \theta \end{bmatrix} + \begin{bmatrix} -0.0435 \\ -0.0043 \\ -0.7662 \\ 0 \end{bmatrix}\begin{bmatrix} \delta_{elev} \end{bmatrix} \tag{1}$$

The longitudinal flight dynamics of the GTM are similar to a conventional aircraft.[11] The dynamics are governed by a slow lightly damped phugoid mode, and a fast lightly damped short period mode. The GTM actuators are modeled independently as 5 Hz first order filters. The time delay in the system, including all communication delay, is 31 msec. Table 1 summarizes the longitudinal dynamics of the GTM.

Table 1. Longitudinal dynamics of the GTM.

Mode	Frequency [rad/s]	Damping	Time Constant [sec]
Phugoid	0.28	0.08	22.44
Short Period	7.10	0.42	0.89
Actuator	31.42		0.2
Time Delay			0.031

The primary control objective is to design a closed-loop system with desirable pilot handling qualities that is robust to time delay and modeling uncertainty. Increased damping in the oscillatory modes is required for satisfactory handling performance. Time domain performance specifications, such as rise times, must also meet pilot expectations. Gain and phase margins must satisfy a minimum of 8 dB and 60 degrees, respectively, to ensure robustness.

Angle-of-attack tracking is selected for the longitudinal axis control architecture, giving the pilot direct command of the desired flight condition. The first feedback to the controller is the error between longitudinal stick command and angle-of-attack. The second feedback signal is the pitch rate. The H_∞ optimization is posed as a model-matching problem, utilizing an ideal tracking model to shape the longitudinal stick command to angle-of-attack response. Figure 2 shows the H_∞ synthesis interconnection for the longitudinal controller. Actuator dynamics and a first order Pade approximation of the time delay are included in the plant GTM_{lon}.

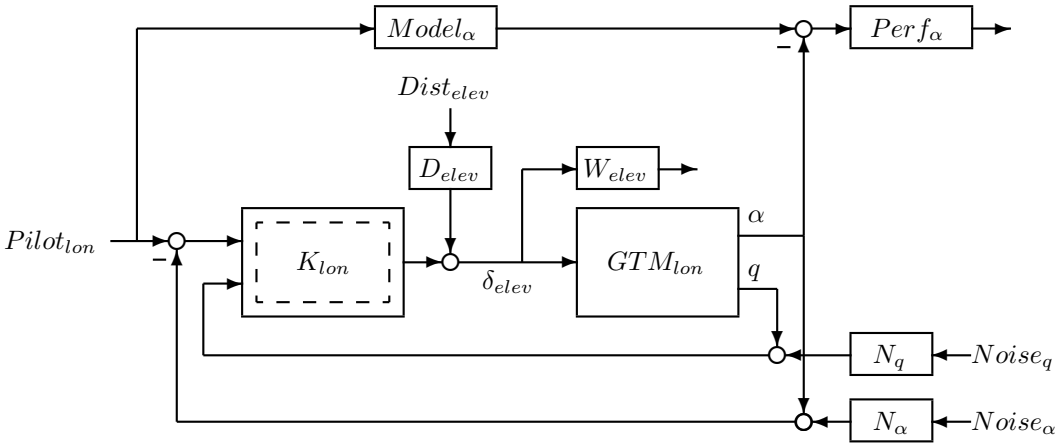


Figure 2. Longitudinal controller synthesis interconnection.

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Ideal models and weighting functions are used by the H_∞ framework to synthesize a controller. The desired input-output response from the pilot command to angle-of-attack is represented by an ideal tracking model. The bandwidth of the tracking model is set to 4 rad/s to achieve a desirable rise time in the closed-loop response. The ideal model is also augmented with actuator dynamics and a linear approximation of the time delay to improve controller performance. A frequency dependent performance weight is used to shape the closed-loop response to match the ideal system. Additional considerations include control effort penalty to avoid high gain in the controller, and measurement noise corruption to prevent plant inversion. Disturbance on the elevator is used to improve robustness. Table 2 summarizes the weighting functions used for the longitudinal H_∞ control synthesis optimization.

Table 2. Weighting functions used to synthesize the longitudinal controller.

Weight	Value
$Model_\alpha$	$\frac{16}{s^2+5.6s+16}(\frac{31.42}{s+31.42})(\frac{-s+64.52}{s+64.52})$
$Perf_\alpha$	$\frac{0.1s+5.973}{s+0.176}$
D_{elev}	0.01
W_{elev}	$\frac{0.8s+42.91}{s+119.2}$
N_q	0.1
N_α	0.01

The performance of the controller is verified through frequency domain analysis. Figure 3 provides the frequency response of the closed-loop system at the design flight condition of 80 knots. The open-loop transfer function from elevator to angle-of-attack is also included, along with the ideal tracking model. The closed-loop transfer function from the pilot command to angle-of-attack is shown as the dashed line in Figure 3. Tracking is achieved up to 4 rad/s as desired. The gain, phase, and time delay margins are 11 dB, 96 degrees, and 722 msec, respectively, which satisfy the specified robustness objectives.

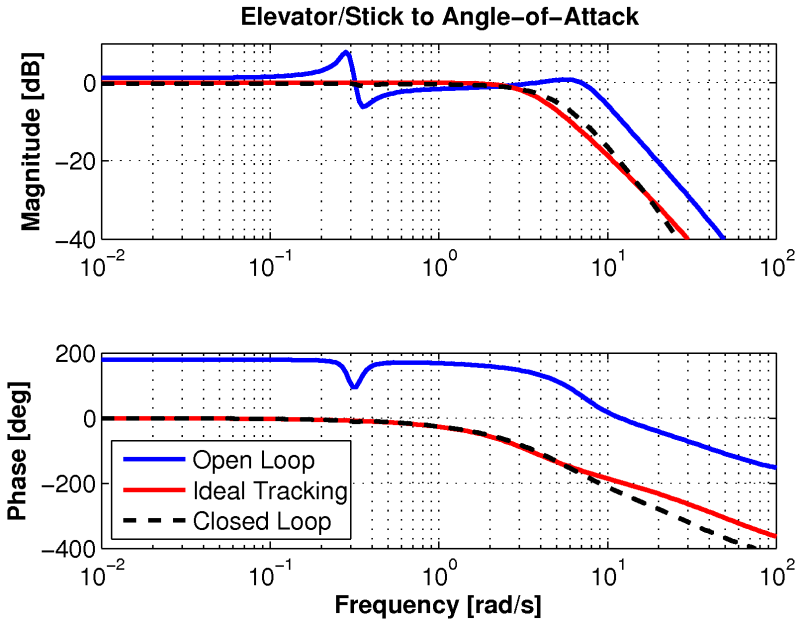


Figure 3. Longitudinal model matching control synthesis.

Stability margins are also verified throughout the full airspeed flight envelope, from 60 to 100 knots. Over the entire flight envelope, the minimum gain margin is 8 dB at 100 knots. The minimum phase margin is 56 degrees at 60 knots, and the minimum time delay margin is 126 msec at 100 knots. These stability margins are sufficiently high to meet the robustness objectives for the closed-loop system over the entire flight envelope.

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Angle-of-attack tracking performance is verified for the full airspeed flight envelope. Performance is evaluated by analyzing the peak deviation in gain between the ideal model and the closed-loop system for all frequencies up to the bandwidth of the closed-loop system. At the design condition, the peak gain deviation is 0.8 dB. Performance degrades most significantly as airspeed decreases due to a resonant peak near the bandwidth of the closed-loop system. This resonant peak is attributed to an increase in the open-loop gain. The worst case performance degradation occurs at 60 knots where the resonant peak has a gain of 5.6 dB near 4 rad/s. Although closed-loop damping is reduced at low airspeed, the general angle-of-attack tracking performance objective is satisfied across the entire flight envelope.

Finally, a stick shaping function is applied to the input command. The shaping function is designed according to pilot requests to calibrate the physical stick deflections. The longitudinal stick shaping function is implemented as a look-up table, and given by the input vector [-1.000 -0.5 -0.25 0.000 0.25 0.5 1.000], and the output vector [30 10 5 0 -1 -5 -10].

B. Lateral/Directional Aircraft Dynamics and Control Design

The states of the lateral/directional model are angle of sideslip β [rad], roll rate p [rad/s], yaw rate r [rad/s], and bank angle ϕ [rad]. The control inputs are aileron and rudder deflections, δ_{ail} [deg] and δ_{rud} [deg], respectively. The lateral/directional state-space model of the GTM at 80 knots is given by the system in Equation 2.

$$\frac{d}{dt}\begin{bmatrix}\beta\\ p\\ r\\ \phi\end{bmatrix} = \begin{bmatrix}-0.5229 & 0.0861 & -0.9852 & 0.2347\\ -92.6425 & -6.2736 & 2.0861 & 0\\ 29.1843 & -0.4833 & -1.4043 & 0\\ 0 & 1 & 0.0857 & 0\end{bmatrix}\begin{bmatrix}\beta\\ p\\ r\\ \phi\end{bmatrix} + \begin{bmatrix}-0.0002 & 0.0031\\ -0.9174 & 0.2321\\ -0.0523 & -0.4436\\ 0 & 0\end{bmatrix}\begin{bmatrix}\delta_{ail}\\ \delta_{rud}\end{bmatrix} \tag{2}$$

The lateral/directional dynamics of the GTM are governed by a very slow stable spiral mode, a fast lightly damped Dutch roll mode, and a slightly faster stable roll mode. Actuator and time delay models as the same as they were for the longitudinal case. Table 3 summarizes the lateral/directional dynamics of the GTM.

Table 3. Lateral/directional dynamics of the GTM.

Mode	Frequency [rad/s]	Damping	Time Constant [sec]
Spiral	0.04		157.08
Dutch Roll	6.25	0.14	1.01
Roll	6.47		0.97
Actuator	31.42		0.2
Time Delay			0.031

The design objectives for the lateral/directional controller are similar to those for the longitudinal controller. The closed-loop system must achieve satisfactory pilot handling performance and be robust with respect to time delay and modeling uncertainty. For the lateral/directional closed-loop system, damping is to be increased in the Dutch roll mode to reduce oscillations.

Stability augmentation alone is selected in the lateral axis since the open-loop dynamics do not require significant modification. Sideslip angle tracking in the directional axis is desired to enable the pilot to perform coordinated turns and crosswind landings. The first feedback to the controller is lateral stick command. The second feedback signal is the error between pedal command and angle of sideslip. The third and fourth feedback signals are roll rate and yaw rate, respectively. An H_∞ model-matching problem is posed for the lateral/directional controller, and the interconnection is shown in Figure 4. Actuator dynamics and a first order Pade approximation of the time delay are included in the plant $GTM_{lat/dir}$.

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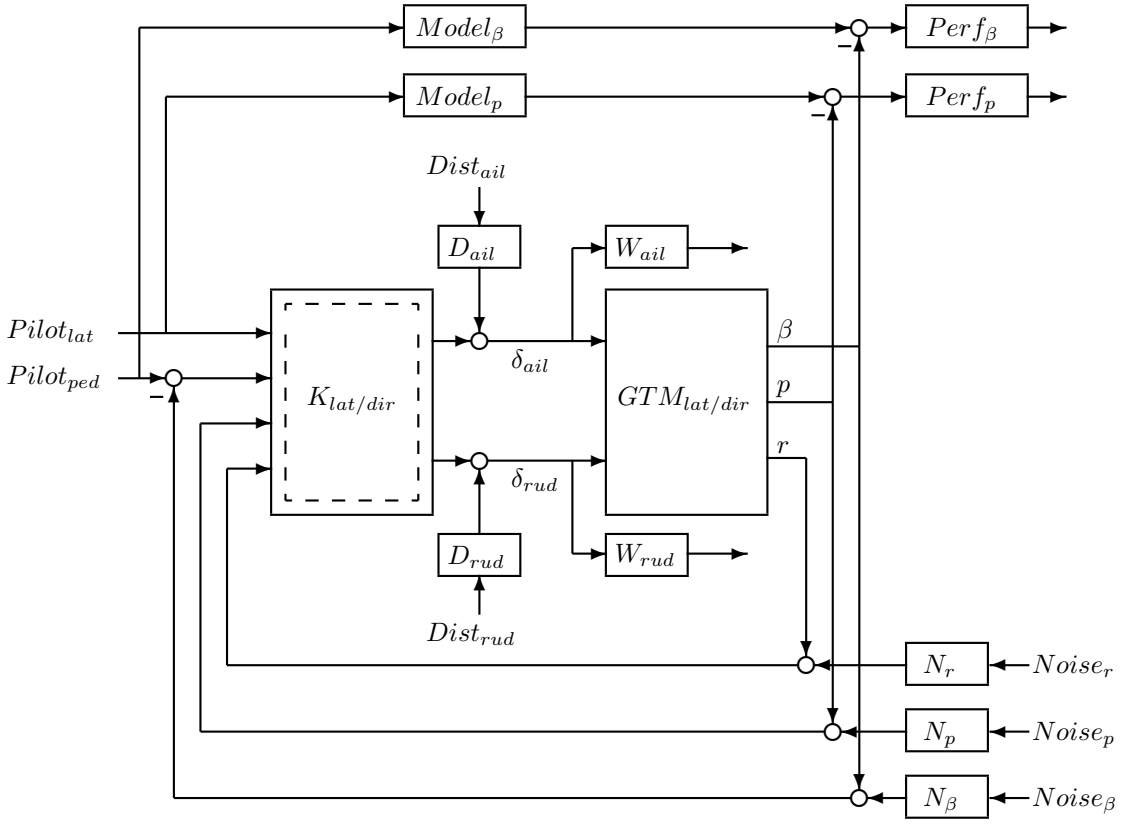


Figure 4. Lateral/directional controller synthesis interconnection.

An ideal model based on stability augmentation is used to shape the lateral stick command to roll rate response. This model is taken as the open-loop transfer function from aileron input to roll rate, with increased damping in the Dutch roll mode. An ideal tracking model is used to shape the pedal command to angle of sideslip response. The bandwidth of the tracking model is set to 2 rad/s to achieve the desired rise time. Both matching models are augmented with actuator and time delay dynamics to improve controller performance. Frequency dependent performance weights are used to achieve the desired closed-loop dynamics. Additionally, control effort penalty, noise corruption, and disturbance injection are included. Table 4 summarizes the weighting functions used for the longitudinal H_∞ control synthesis optimization.

Table 4. Weighting functions used to synthesize the lateral/directional controller.

Weight	Value
$Model_\beta$	$\frac{7.78}{s^2+3.92s+7.78}(\frac{31.42}{s+31.42})(\frac{-s+64.52}{s+64.52})$
$Model_p$	$\frac{-52.56(s-0.02)(s^2+7.5s+34.67)}{(s+0.04)(s+6.47)(s^2+8.75s+39.06)}(\frac{31.42}{s+31.42})(\frac{-s+64.52}{s+64.52})$
$Perf_\beta$	$\frac{0.1s+5.973}{s+0.176}$
$Perf_p$	$\frac{0.1s+5.973}{s+0.299}$
D_{ail}	0.01
D_{rud}	0.01
W_{ail}	$\frac{0.8s+42.91}{s+119.2}$
W_{rud}	$\frac{0.8s+42.91}{s+119.2}$
N_r	0.1
N_p	0.1
N_β	0.01

Controller performance is verified through frequency domain analysis. Figure 5 provides the frequency response of the closed-loop system at the design flight condition of 80 knots. The open-loop transfer functions

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from aileron and rudder inputs to angle of sideslip and roll rate are included, along with the ideal models. The closed-loop transfer functions from commands to responses are shown as dashed lines in Figure 5. Angle of sideslip tracking is achieved up to 2 rad/s as desired. The closed-loop lateral dynamics achieve higher damping in the Dutch roll mode, but otherwise are similar to the open-loop response. The cross-coupling between the inputs and the outputs is also greatly reduced by the controller. The gain, phase, and time delay margins are 14 dB, 75 degrees, and 958 msec, respectively, which satisfy the specified robustness objectives.

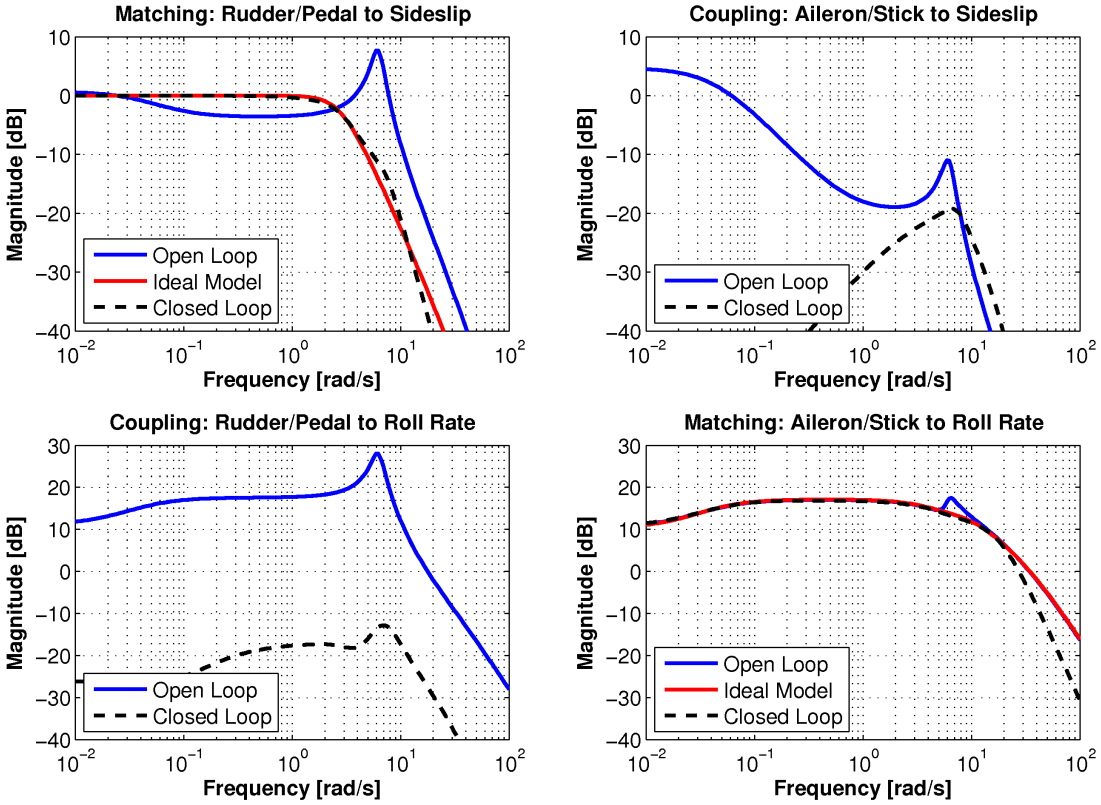


Figure 5. Lateral/directional model matching control synthesis.

Stability margins are also verified throughout the full airspeed flight envelope. A minimum gain margin of 3 dB occurs at 60 knots. The minimum phase margin is 74 degrees, and the minimum time delay margin is 550 msec, both at 100 knots. Although the gain margin is significantly degraded at 60 knots, the overall stability margins of the closed-loop systems are sufficiently high to meet the specified robustness objectives.

Angle of sideslip tracking and lateral axis performance are verified over the full airspeed flight envelope. At the design condition, peak gain deviation between ideal models and the closed-loop system are 0.2 dB in the pedal to sideslip channel, and 0.3 dB in the stick to roll rate channel. The peak cross-coupling gain is -19 dB from stick to sideslip, and -13 dB from rudder to roll rate. In general, performance degrades most significantly for lower airspeed. Sideslip tracking performance, however, is not sensitive changes in airspeed. The worst case gain deviation is 0.5 dB at 60 knots. The closed-loop stick to roll rate response alters to match the changing open-loop dynamics. However, the changes to the open-loop dynamics are benign, hence the performance of the controller is not affected significantly. The peak cross-coupling gain increases to -9 dB from stick to sideslip, and to -12 dB from pedal to roll rate. Although the performance is slightly degraded and cross-coupling increases over the full flight envelope, the overall control objectives are satisfied.

Finally, a stick shaping function is applied to the lateral input command. The shaping function is designed according to pilot requests to calibrate the physical stick deflections. The lateral stick shaping function is given by $Pilot_{lat} = 20u\,|u|^{1.75}$, where u corresponds to the actual pilot input.

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C. Time Domain Verification

The final step prior to flight testing is to verify the performance of the controller using time domain simulations. Linear and nonlinear time domain simulations provide several valuable insights. Simulations can be used to determine the validity of the linear approximation for the aircraft dynamics. If the linear and nonlinear closed-loop responses vary significantly from each other, linear control design and the use of linear analysis tools may not be applicable. Simulations can also be used in conjunction with frequency domain analysis to specifically verify that the controller satisfies the performance objectives. The linear and nonlinear verification simulations are performed at the design flight condition of 80 knots.

Flight tests are ultimately used evaluate the performance of the controller on the real aircraft. The flight maneuvers are driven by automatically generated doublet inputs on the pilot commands. The input signals were designed by NASA flight test engineers to evaluate the longitudinal and lateral performance of control laws for the GTM. Figure 6 shows an example of the input signals for longitudinal and lateral doublets and accounts for the stick shaping functions. Note that the longitudinal stick shaping function is not symmetric, resulting in the asymmetric input seen in Figure 6. Hence, the angle-of-attack and pitch rate responses for the longitudinal doublet will also not be symmetric.

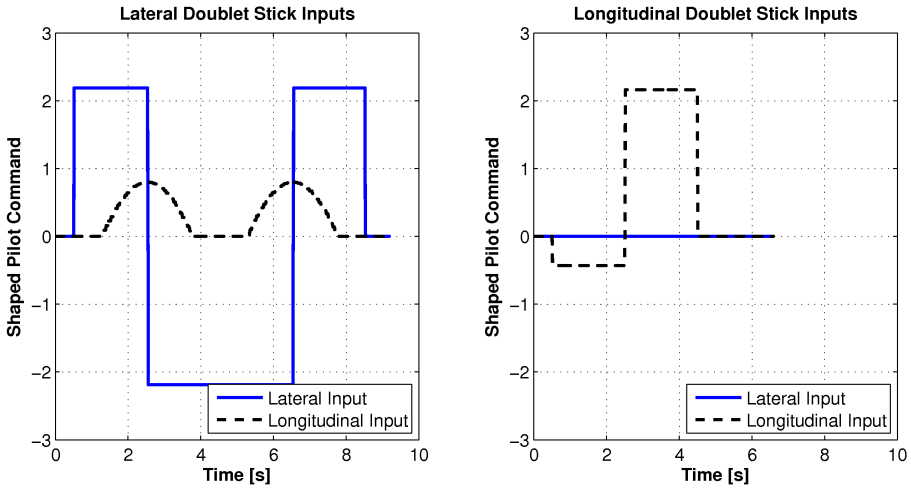


Figure 6. Automatically generated lateral and longitudinal doublet wavetrains.

The input signals shown in Figure 6 are used for the linear and nonlinear simulations to verify the expected performance of the controller prior to flight testing. Doublet responses are shown in Figure 7, with linear and nonlinear simulation results shown together. The lateral doublet state response is shown in the left column, and the longitudinal doublet state response is shown in the right column.

Time domain simulations indicate that the linear approximation for the GTM flight dynamics is valid. With the exception of small deviations in the pitch rate during the lateral doublet, the closed-loop response in the nonlinear simulation is nearly identical to its linear counterpart. The results in Figure 7 therefore validate the use of linear control design techniques and analysis tools for this particular flight condition. The results also confirm that the control law satisfies the performance objectives. The lateral response during the doublet maneuver exhibits no oscillation. The angle-of-attack is tracked according to the ideal model during the longitudinal doublet. Hence, the performance of the overall control architecture and its satisfaction of the control objectives is verified.

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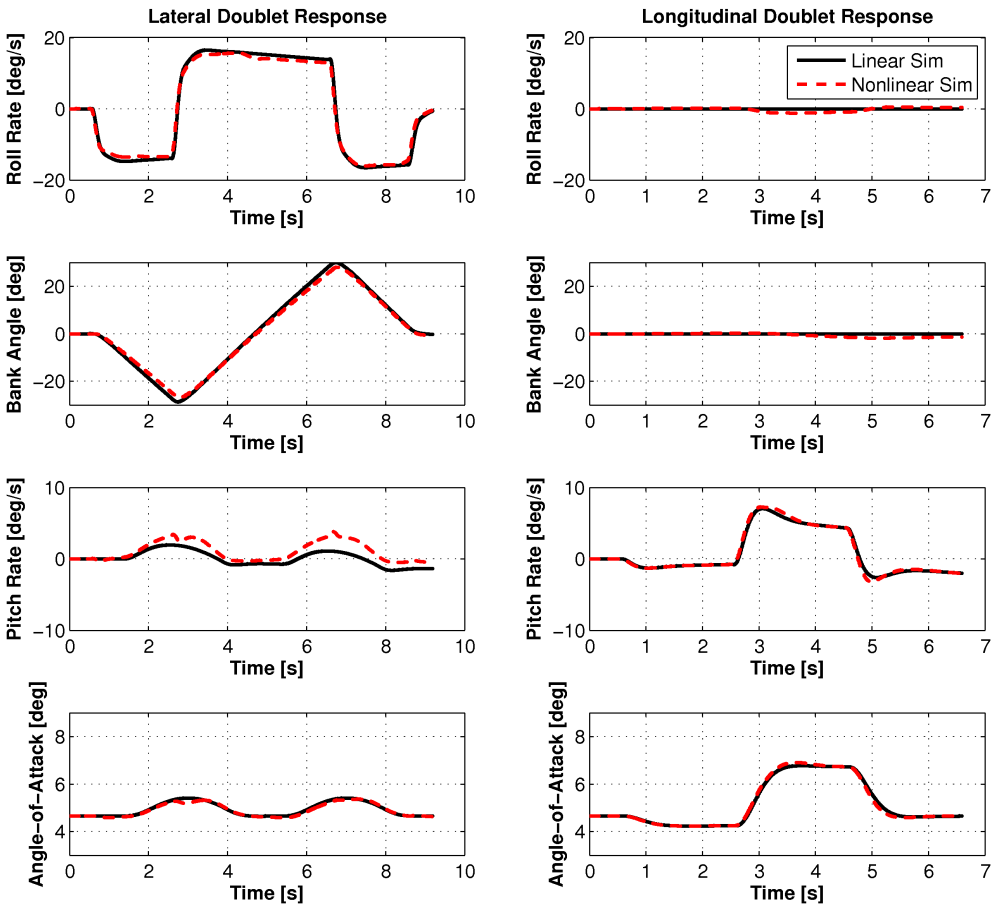


Figure 7. Doublet simulation responses in lateral and longitudinal axes for verification.

III. Flight Test Results

The control law was flight tested on the AirSTAR GTM T-2 aircraft in September, 2010 at Fort Pickett, Virginia. Performance and robustness properties of the closed-loop system were evaluated for a nominal flight condition and for various adverse scenarios. The adverse scenarios comprised of an excess time delay test and a stability degradation test. Details on the nominal flight test and results are presented in Subsection A, the excess time delay test and results in Subsection B, and the stability degradation test and results in Subsection C.

The target flight condition for the various flight tests was an airspeed of 80 knots and an altitude around 1000 feet. Automatically generated longitudinal and lateral doublet inputs were applied during the flight tests to evaluate performance and robustness properties of the closed-loop system. In addition to doublet inputs, offset landing maneuvers were performed under the stability degradation scenario to specifically evaluate pilot handling qualities.[12] Pilot interviews were conducted immediately following each offset landing maneuver to determine Cooper-Harper ratings.[13]

A. Nominal Aircraft Performance

Nominal performance of the closed-loop system was evaluated using a series of longitudinal and lateral doublets. Doublet responses are shown in Figure 8, with nonlinear simulation prediction and actual flight test results shown together. The lateral doublet state response is shown in the left column, and the longitudinal doublet state response is shown in the right column. The nonlinear simulation results were obtained by trimming the simulation model to the same flight condition as the flight experiment and using the recorded pilot commands as inputs for the maneuvers.

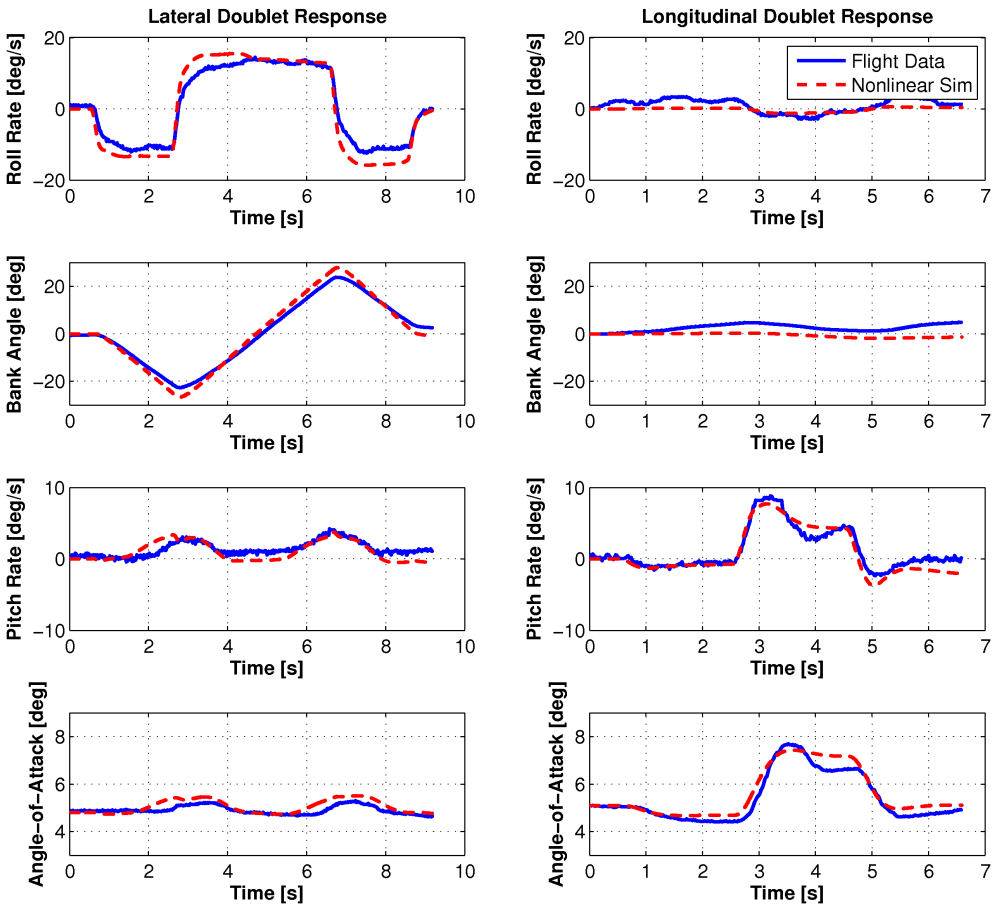


Figure 8. Nominal doublet response in lateral and longitudinal axes.

The nonlinear GTM simulation accurately predicts the in-flight response of the lateral states. In particular, the bank angle prediction is very accurate when compared to the flight response during the lateral doublet. Comparing the roll rate responses reveals a small discrepancy in the system gain. The discrepancy suggests that the aircraft has less gain in this channel than predicted by the simulation model. The control objective of reducing Dutch roll mode oscillations through a stability augmentation system is achieved. The performance of the lateral component of the controller is thus validated by the flight test.

The longitudinal doublet response is also captured well by the nonlinear simulation. The predicted gain in the angle-of-attack and pitch rate channels is accurate. However, the in-flight doublet response exhibits less damping than predicted by the nonlinear simulation. The angle-of-attack command is tracked with no oscillation in the nonlinear simulation, which was the objective of the longitudinal control law. The in-flight performance degradation is attributed to errors in the aerodynamic model.

The doublet maneuvers executed on the nominal system are not only useful in validating the performance of the control law, but also in validating the accuracy of the simulation model itself. The controller is designed based on the assumption that the dynamics of the simulation model represent the true aircraft. Hence, it is important to analyze the accuracy of the simulation model. Based on the results in Figure 8, the simulation model is shown to accurately capture the general behavior of the aircraft. Slight discrepancies are noted in the roll gain and the pitch damping. These discrepancies are expected to play a role in validating the performance and robustness of the controller during the adverse scenario flight tests.

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B. Performance and Robustness Subject to Excess Time Delay

An excess time delay test was performed in an attempt to identify the time delay margin of the closed-loop system. AirSTAR control laws are implemented on a ground-based flight computer, and there is an inherent 31 msec time delay in the control loop due to telemetry and computation delay. Additional time delay was introduced directly on the actuator commands during the failure flight test to assess the robustness of the controller to time delay. While the research pilot flew racetrack circuits, increments of 5 msec time delay were added every 5 seconds to the actuator commands. On each straightaway of the circuit, the pilot executed a longitudinal or a lateral doublet. The incremental time delay test was to be halted when the aircraft becomes unstable or uncontrollable.

A sample of the excess time delay test results is presented in Figure 9. The nonlinear simulation prediction and the flight test results are shown together. The lateral doublet state response at 65 msec of extra time delay is shown in the left column. The right column shows a longitudinal doublet at 105 msec of extra time delay. Performance degradation is noted through small oscillations, which is particularly visible in the roll rate measurement. Similarly, damping in the longitudinal axis is reduced, which is visible in the pitch rate of the nonlinear simulation prediction.

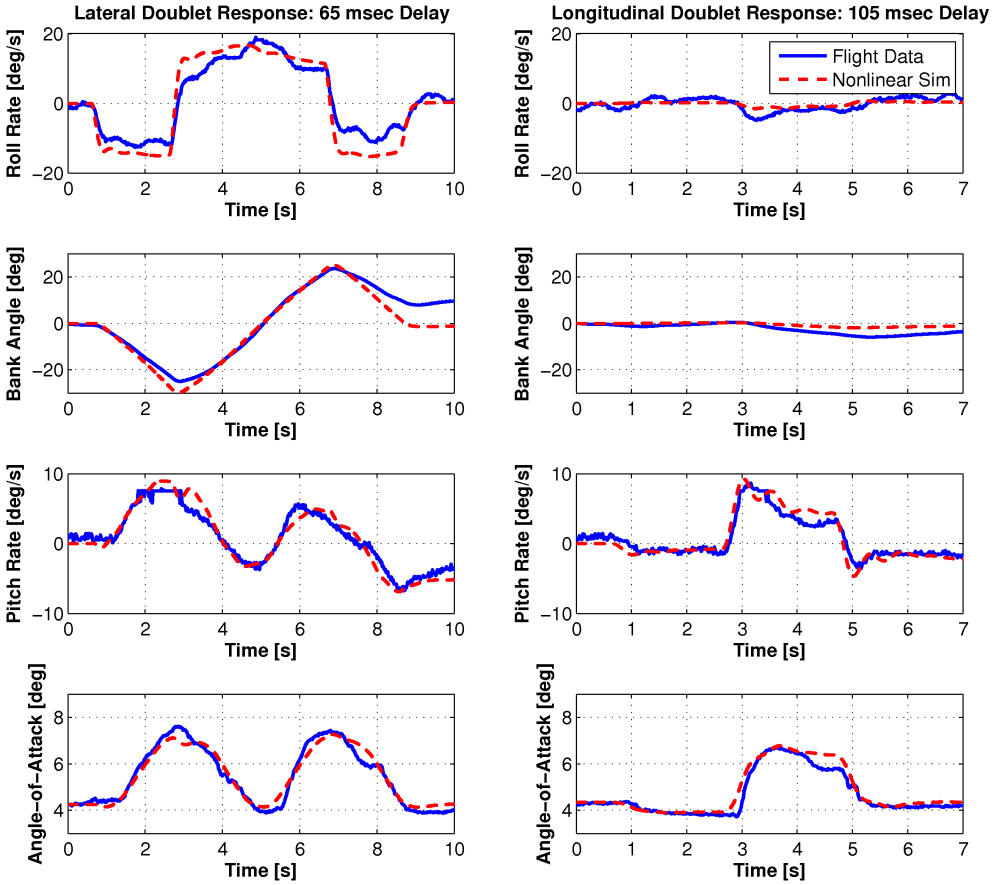


Figure 9. Excess time delay doublet response in lateral and longitudinal axes.

The excess time delay flight test was halted immediately following a turn to align the aircraft for a subsequent doublet maneuver. The extra time delay in the control loop when the test was halted was 185 msec. Pilot comments indicate that handling qualities in the longitudinal axis had become unacceptable during the turn. The angle-of-attack response during a turn of the nominal aircraft is shown on the left in Figure 10. The pilot longitudinal stick command during the same turn is included below. The angle-of-attack response during the turn after which the test was halted, i.e. with 185 msec of extra time delay in the loop, is shown on the right. The pilot longitudinal stick command is again included below. A significant increase in oscillations is noted due to the extra time delay. The nonlinear simulation results were obtained

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by using the recorded pilot stick commands as inputs to the simulation model. Similar to the nominal doublet response comparison in Figure 8, the in-flight angle-of-attack response exhibits less damping than predicted by the simulation model. The discrepancy in damping between flight test and simulation is noted during the nominal turn as well as during the turn with excess time delay. Simulation suggests that although oscillations are more severe due to the 185 msec of excess time delay, the closed-loop system remains stable in the presence of further time delays.

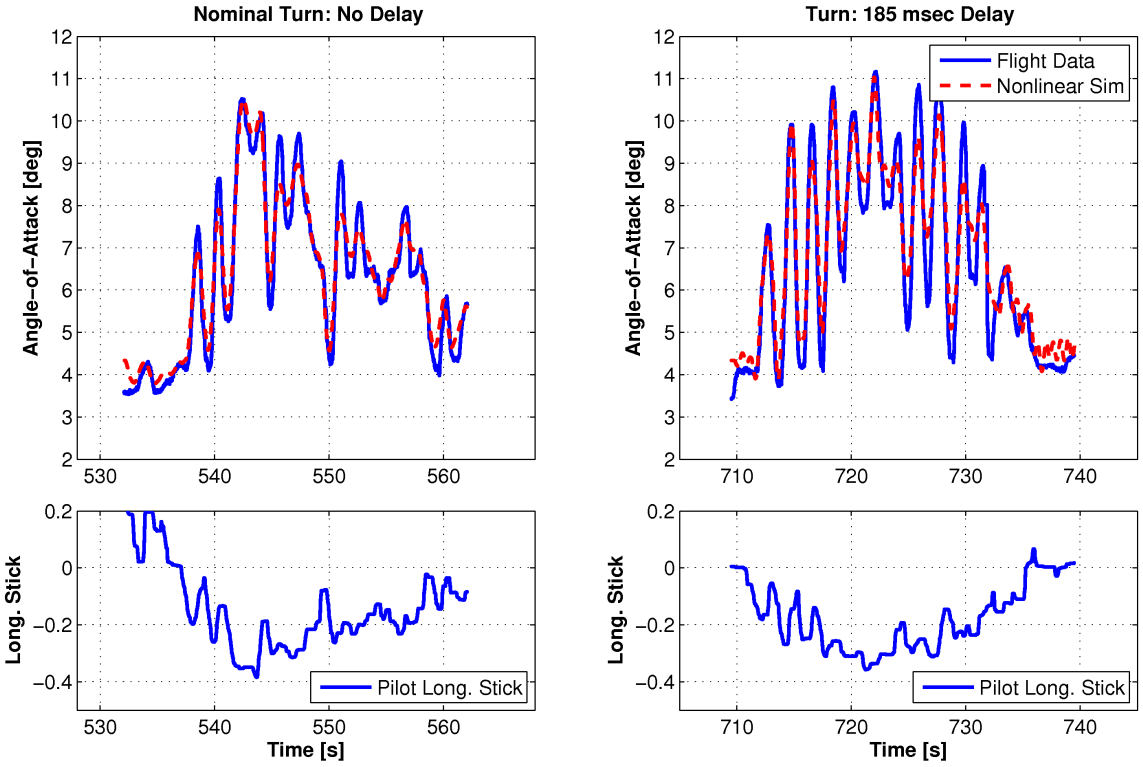


Figure 10. Excess time delay leads to reduced longitudinal axis damping.

The results of the excess time delay flight test provide several crucial insights. Most importantly, the results indicate that the time delay margin may not always be a relevant metric for evaluating flight control laws. The time delay margin for the target flight condition, computed using linearized aircraft dynamics, is above 700 msec. Nonlinear simulations are stable in excess of 600 msec of time delay in the loop. However, the pilot felt that handling qualities were unsatisfactory with just 185 msec of extra time delay. Therefore, there is a clear distinction between handling qualities and closed-loop stability. A metric aimed at evaluating the robustness of the performance of the closed loop could be more insightful. A second important point is to note that the excess time delay did not lead to Pilot Induced Oscillations (PIOs). The pilot longitudinal stick commands are included in Figure 10 to show specifically that the adverse response is not due to a PIO. There is no qualitative difference in the pilot longitudinal stick command when comparing the nominal turn to the turn with 185 msec of extra time delay.

C. Performance and Robustness Subject to Stability Degradation

Stability degradation tests were performed to verify the robustness of the controller to model uncertainty. Redundant surfaces on the GTM were used to degrade the longitudinal and lateral stability of the aircraft using inner-loop feedback. Feedback of the angle-of-attack to the inboard elevator segment emulates longitudinal static stability degradation. This failure is equivalent to reducing the pitch stiffness C_{m_α}, where 100% reduction represents neutral static stability. Feedback of the roll rate to the spoilers emulates lateral dynamic stability degradation. This failure is designed to decrease the roll damping term C_{l_p}, but the feedback of the roll rate also has effects in the directional axis due to coupling.

Simultaneous stability degradation failures in the longitudinal and lateral axes were tested with doublets

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and offset landing maneuvers. Cooper-Harper ratings were calculated during the offset landing maneuvers based on pilot comments regarding the handling properties of the aircraft. Table 5 summarizes the Cooper-Harper ratings for the nominal case and for various levels of stability degradation as found during the offset landing maneuvers. The nominal handling performance of the controller was rated in the 'satisfactory without improvement' category on the Cooper-Harper scale.

Table 5. Cooper-Harper ratings evaluated during offset landing maneuvers.

Test	Degradation	C-H Rating	Handling
1	Nominal	3	Some mildly unpleasant deficiencies
2	100% C_{m_α}, +0.4 C_{l_p}	5	Moderately objectionable deficiencies
3	125% C_{m_α}, +0.5 C_{l_p}	10	Major deficiencies (Unstable)

Longitudinal and lateral doublets maneuvers were also conducted for various levels of stability degradation. Table 6 summarizes the results of these flight tests. The flight test results indicate that the longitudinal axis remained stable for the maximum degradation in pitch stiffness. However, the lateral/directional closed-loop dynamics became unstable with the degradation corresponding to Test 3 in Table 6. Lateral doublets were not attempted for stability degradations corresponding to Tests 4 and 5.

Table 6. Stability evaluated with doublet maneuvers.

Test	Degradation	Stable Pitch	Stable Roll
1	None	Yes	Yes
2	50% C_{m_α}, +0.2 C_{l_p}	Yes	Yes
3	75% C_{m_α}, +0.3 C_{l_p}	Yes	No
4	100% C_{m_α}, +0.4 C_{l_p}	Yes	N/A
5	125% C_{m_α}, +0.5 C_{l_p}	Yes	N/A

IV. Conclusion

The successful implementation of a robust multivariable controller on the NASA AirSTAR GTM represents a significant milestone. The design approach was validated by the results of various rigorous performance and robustness flight tests. Excessive time delay and stability degradation in the aircraft dynamics were tolerated by the closed-loop system. Additionally, the controller performed well during offset landings.

A controller re-design is necessary to enhance performance and robustness properties. The test pilot gave the closed-loop system a Cooper-Harper rating of 3 for the nominal case. Although this is a satisfactory grade on a first iteration, better handling qualities are desired. Further, an improved balance of robustness properties will allow the controller to tolerate higher degrees of stability degradation. Such a re-design would enable the controller to become a candidate for the permanent baseline control law for the GTM. The lessons learned from the GTM control design process also present an opportunity to design an even better controller for the next generation AirSTAR flight test vehicle.

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